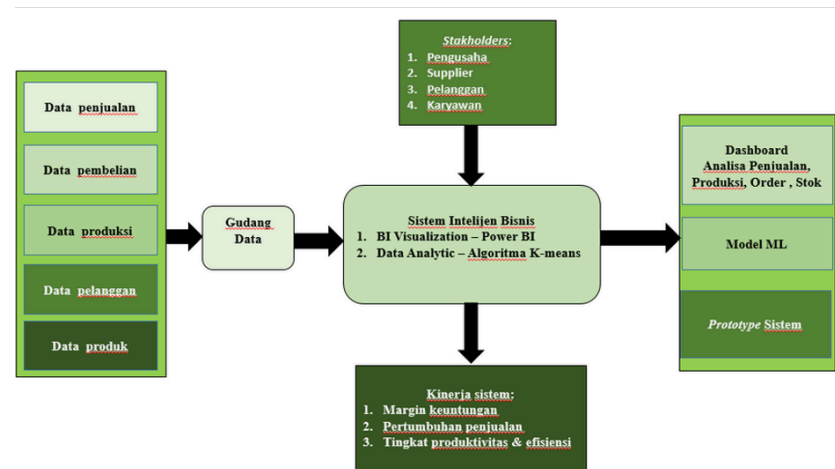
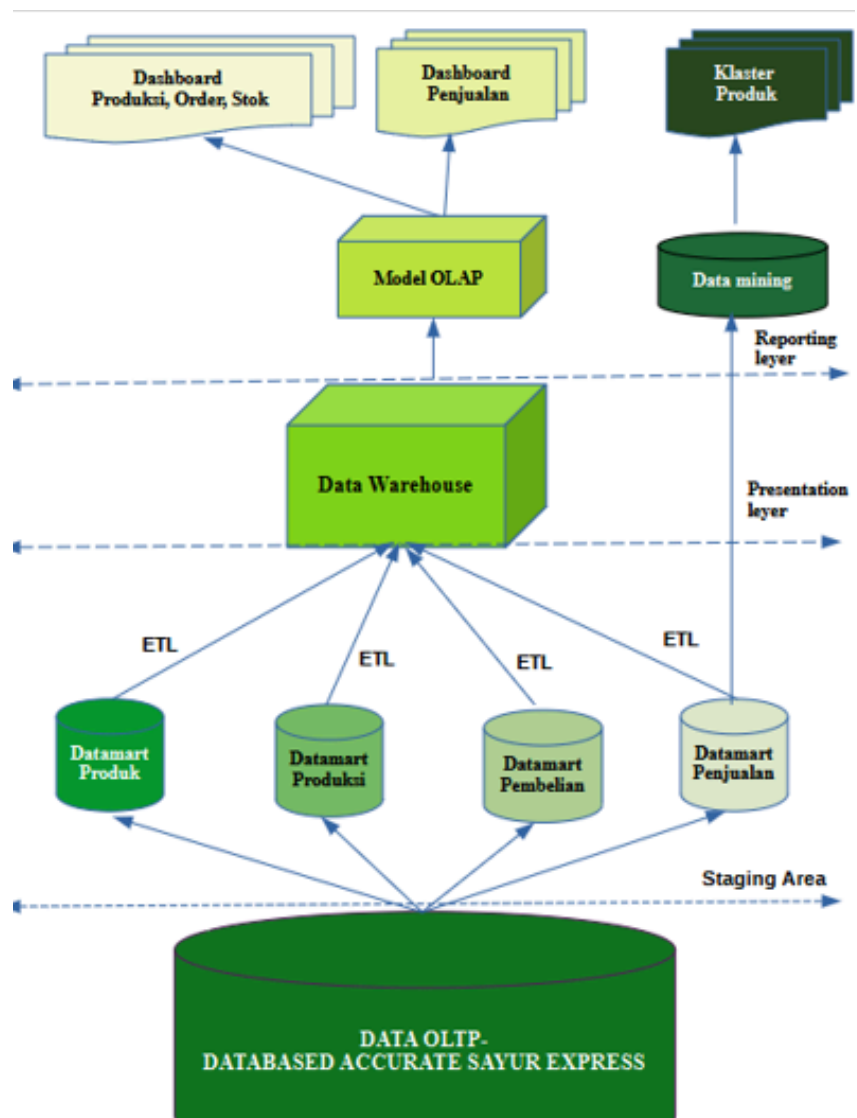


1. Design model sistem intelijen bisnis yang dikembangkan



2. Arsitektur data pipeline



### 3. Source data Online Order dari API marketplace

**Tabel Order Online**

| #  | Name                       | Type          |
|----|----------------------------|---------------|
| 1  | Order_ID 📄                 | varchar(15)   |
| 2  | Order_Number               | varchar(15)   |
| 3  | Product_Name               | varchar(100)  |
| 4  | Category_Code              | varchar(10)   |
| 5  | Product_ID 📄               | varchar(15)   |
| 6  | Order_Status               | varchar(15)   |
| 7  | Pick_up_Status             | varchar(15)   |
| 8  | Date_of_Make_Order         | date          |
| 9  | Product_Price              | decimal(10,2) |
| 10 | Discount_Price             | decimal(10,2) |
| 11 | Number                     | decimal(10,2) |
| 12 | Returned_quantity          | decimal(10,2) |
| 13 | Total_Product_Price        | decimal(10,2) |
| 14 | Discount_Total             | decimal(10,2) |
| 15 | Discount_by_Seller         | decimal(10,2) |
| 16 | Discount_by_Shopee         | decimal(10,2) |
| 17 | Order_Quantity             | decimal(10,2) |
| 18 | Voucher_by_Seller          | decimal(10,2) |
| 19 | Cashback_Coin              | decimal(10,2) |
| 20 | Voucher_by_Shopee          | decimal(10,2) |
| 21 | Discount_Package           | decimal(10,2) |
| 22 | Discount_Package_by_Shopee | decimal(10,2) |
| 23 | Discount_Package_by_Seller | decimal(10,2) |
| 24 | Discount_Coin_by_Shopee    | decimal(10,2) |
| 25 | Discount_by_Credit_Card    | decimal(10,2) |
| 26 | Delivery_Cost_by_Seller    | decimal(10,2) |
| 27 | Discount_of_Delivery_Cost  | decimal(10,2) |
| 28 | Total_Payment              | decimal(10,2) |
| 29 | Estimation_Delivery_Cost   | decimal(10,2) |
| 30 | Product_Size               | decimal(10,2) |
| 31 | Total_Product_Size         | decimal(10,2) |
| 32 | Unit                       | varchar(5)    |
| 33 | Customer_ID 📄              | varchar(10)   |
| 34 | Code_Branch                | varchar(10)   |

### 4. Data understanding dan pre-processing dengan GoogleColab & Rapidminer

#### Data Shape

```
[12] 1 df_Order = pd.read_excel('/content/Online Order 1-30 Mei 2020.xlsx')
```

```
1 df_Order.shape
```

```
(72533, 50)
```

#### Seleksi Data--> Finished & On Delivery ---> SGP & SGR

| Order Status     | count |
|------------------|-------|
| Finished         | 63005 |
| Canceled         | 6605  |
| On Delivery      | 2694  |
| Needs to be Sent | 229   |

```
[15] 1 df_sales = pd.read_excel('/content/df_order2.xlsx')
```

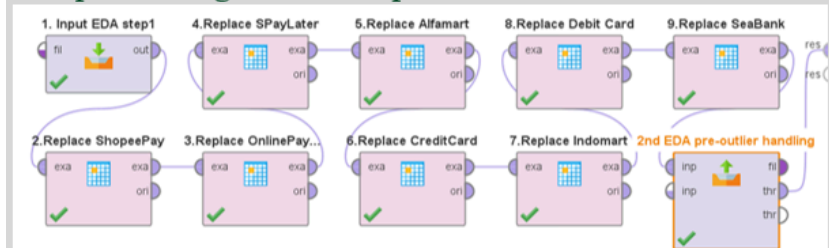
```
1 df_sales.shape
```

```
(40916, 12)
```

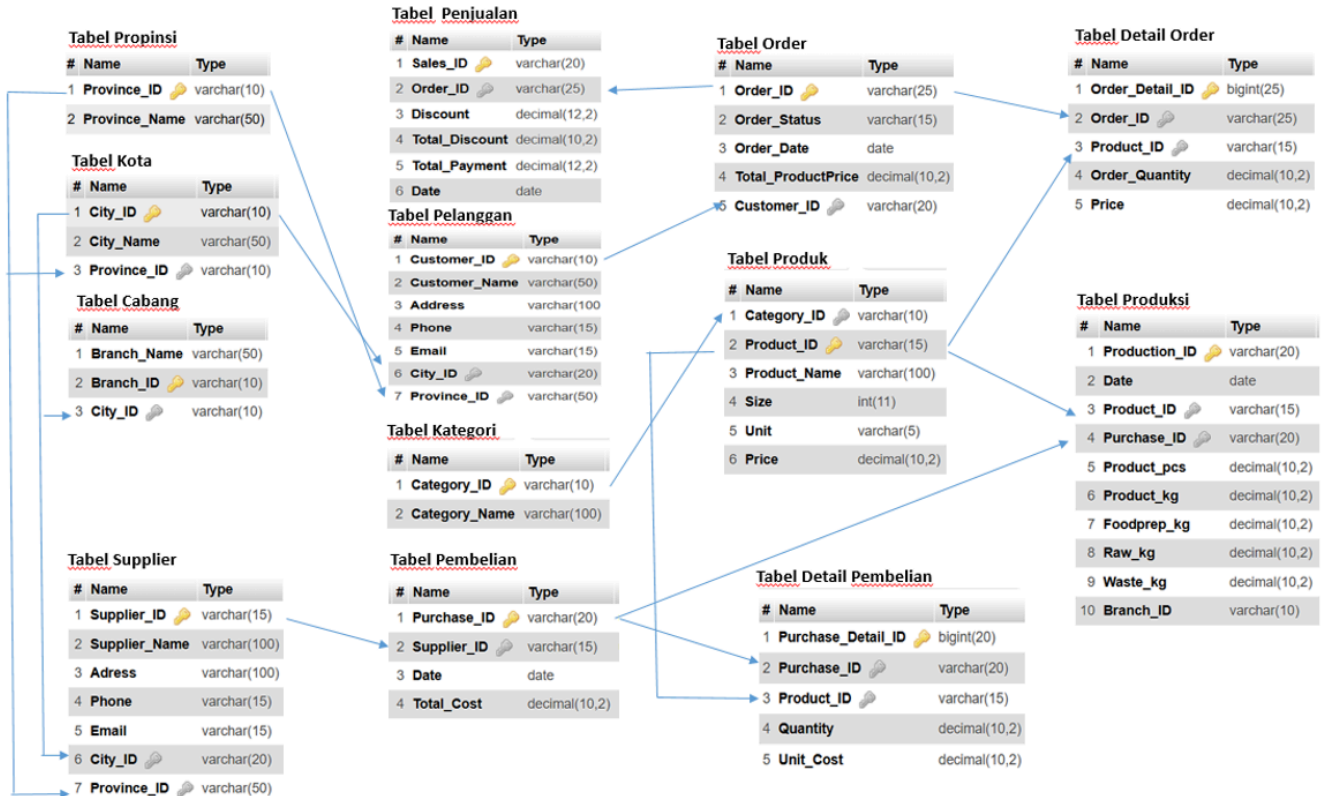
## Pre-processing 1 --- > Filter, Split, Replace, Rename



## Pre-processing 2 --- > Hapus Outlier - > 21.178 baris



## 5. Normalisasi databased dan design RDBMS (MySQL dan phpMyAdmin)



phpMyAdmin

Server: 127.0.0.1 Database: sayur\_express

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers

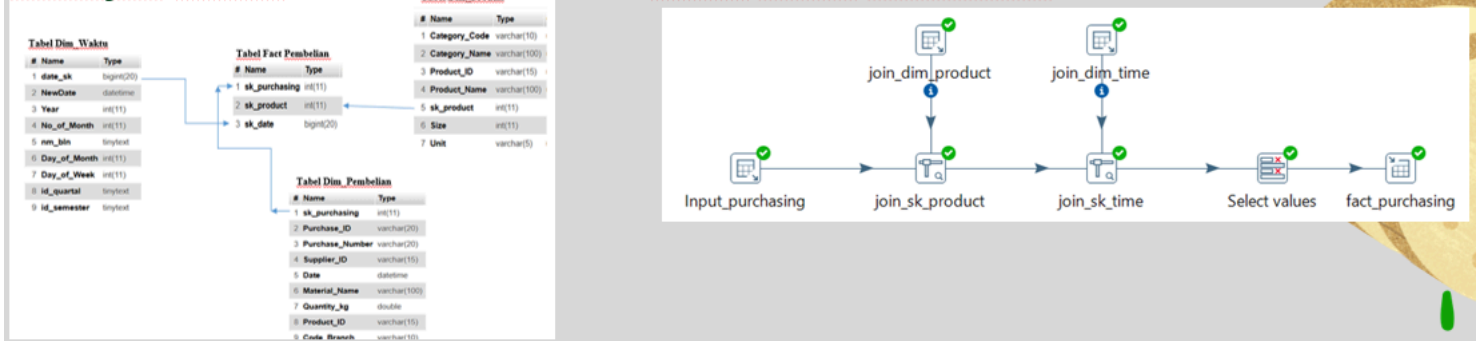
Filters

Containing the word:

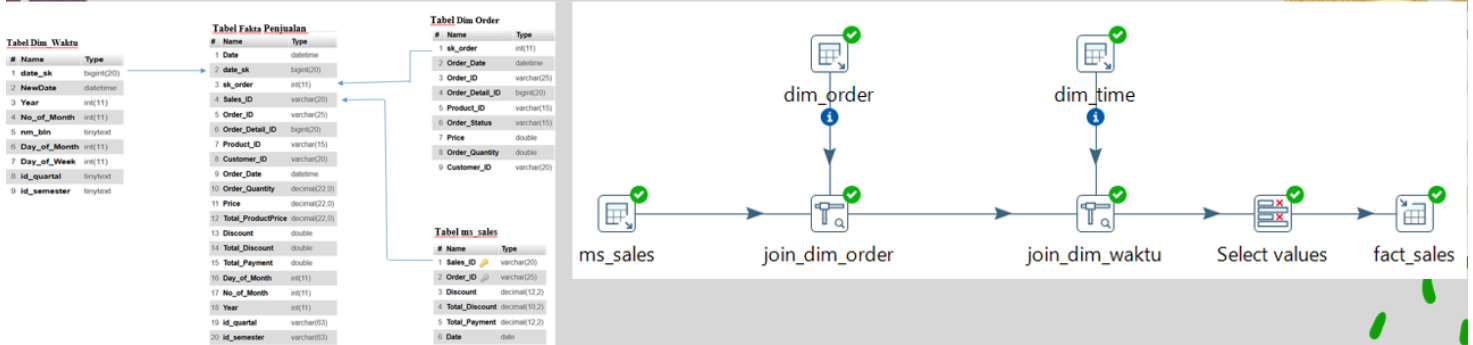
| Table                                        | Action                                      | Rows  | Type   | Collation          | Size      | Overhead |
|----------------------------------------------|---------------------------------------------|-------|--------|--------------------|-----------|----------|
| <input type="checkbox"/> dim_branch          | ★ Browse Structure Search Insert Empty Drop | 5     | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> dim_order           | ★ Browse Structure Search Insert Empty Drop | 5,914 | InnoDB | utf8mb4_general_ci | 1.5 MiB   | -        |
| <input type="checkbox"/> dim_product         | ★ Browse Structure Search Insert Empty Drop | 6     | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> dim_production      | ★ Browse Structure Search Insert Empty Drop | 180   | InnoDB | utf8mb4_general_ci | 80.0 KiB  | -        |
| <input type="checkbox"/> dim_purchasing      | ★ Browse Structure Search Insert Empty Drop | 180   | InnoDB | utf8mb4_general_ci | 64.0 KiB  | -        |
| <input type="checkbox"/> dim_supplier        | ★ Browse Structure Search Insert Empty Drop | 1     | InnoDB | utf8mb4_general_ci | 16.0 KiB  | -        |
| <input type="checkbox"/> dim_time            | ★ Browse Structure Search Insert Empty Drop | 3,600 | InnoDB | utf8mb4_general_ci | 272.0 KiB | -        |
| <input type="checkbox"/> fact_penjualan      | ★ Browse Structure Search Insert Empty Drop | 5,914 | InnoDB | utf8mb4_general_ci | 2.0 MiB   | -        |
| <input type="checkbox"/> fact_produk         | ★ Browse Structure Search Insert Empty Drop | 180   | InnoDB | utf8mb4_general_ci | 80.0 KiB  | -        |
| <input type="checkbox"/> fact_proses         | ★ Browse Structure Search Insert Empty Drop | 180   | InnoDB | utf8mb4_general_ci | 64.0 KiB  | -        |
| <input type="checkbox"/> fact_sisapersediaan | ★ Browse Structure Search Insert Empty Drop | 180   | InnoDB | utf8mb4_general_ci | 80.0 KiB  | -        |
| <input type="checkbox"/> fac_pembelian       | ★ Browse Structure Search Insert Empty Drop | 180   | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> ms_branch           | ★ Browse Structure Search Insert Empty Drop | 5     | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> ms_category         | ★ Browse Structure Search Insert Empty Drop | 5     | InnoDB | utf8mb4_general_ci | 16.0 KiB  | -        |
| <input type="checkbox"/> ms_city             | ★ Browse Structure Search Insert Empty Drop | 7     | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> ms_customer         | ★ Browse Structure Search Insert Empty Drop | 12    | InnoDB | utf8mb4_general_ci | 48.0 KiB  | -        |
| <input type="checkbox"/> ms_order            | ★ Browse Structure Search Insert Empty Drop | 5,914 | InnoDB | utf8mb4_general_ci | 1.8 MiB   | -        |
| <input type="checkbox"/> ms_order_details    | ★ Browse Structure Search Insert Empty Drop | 5,914 | InnoDB | utf8mb4_general_ci | 1.0 MiB   | -        |
| <input type="checkbox"/> ms_product          | ★ Browse Structure Search Insert Empty Drop | 6     | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> ms_production       | ★ Browse Structure Search Insert Empty Drop | 180   | InnoDB | utf8mb4_general_ci | 80.0 KiB  | -        |

## 6. Proses ekstraksi dan transformasi data dengan Pentaho Data Intergration

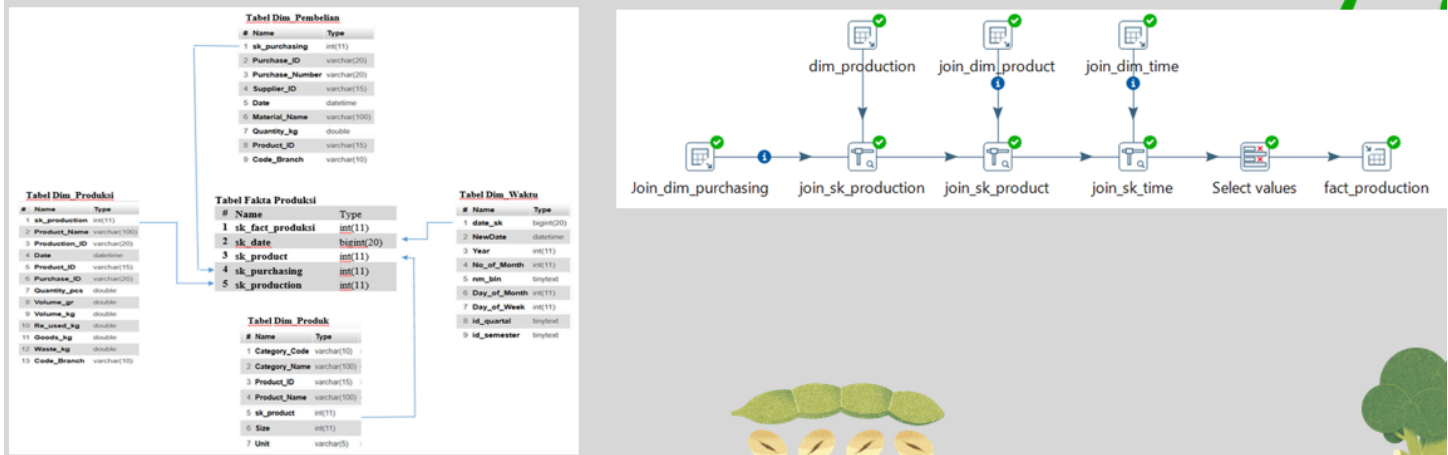
### Intelijen Bisnis -> ETL --> OLAP --- >Tabel Fakta Pembelian



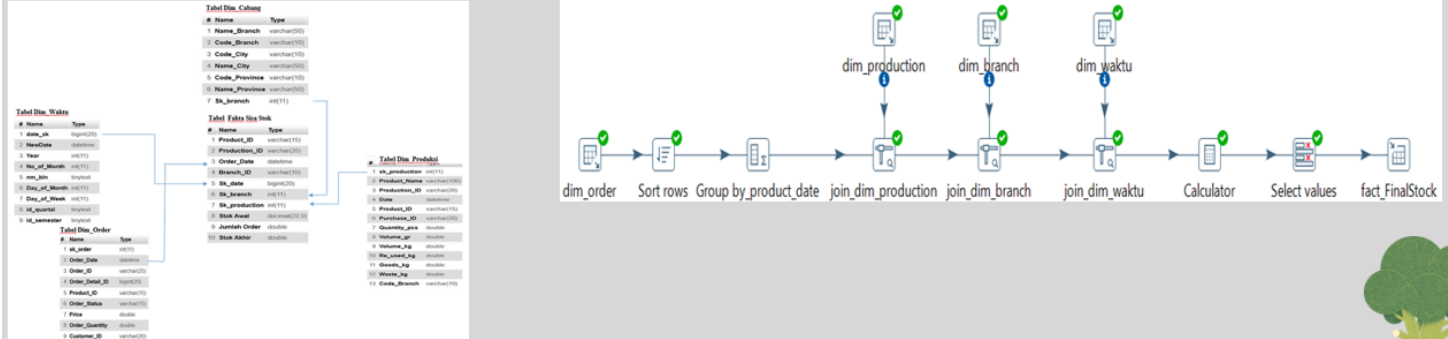
### Intelijen Bisnis -> ETL --> OLAP --- >Tabel Fakta Penjualan



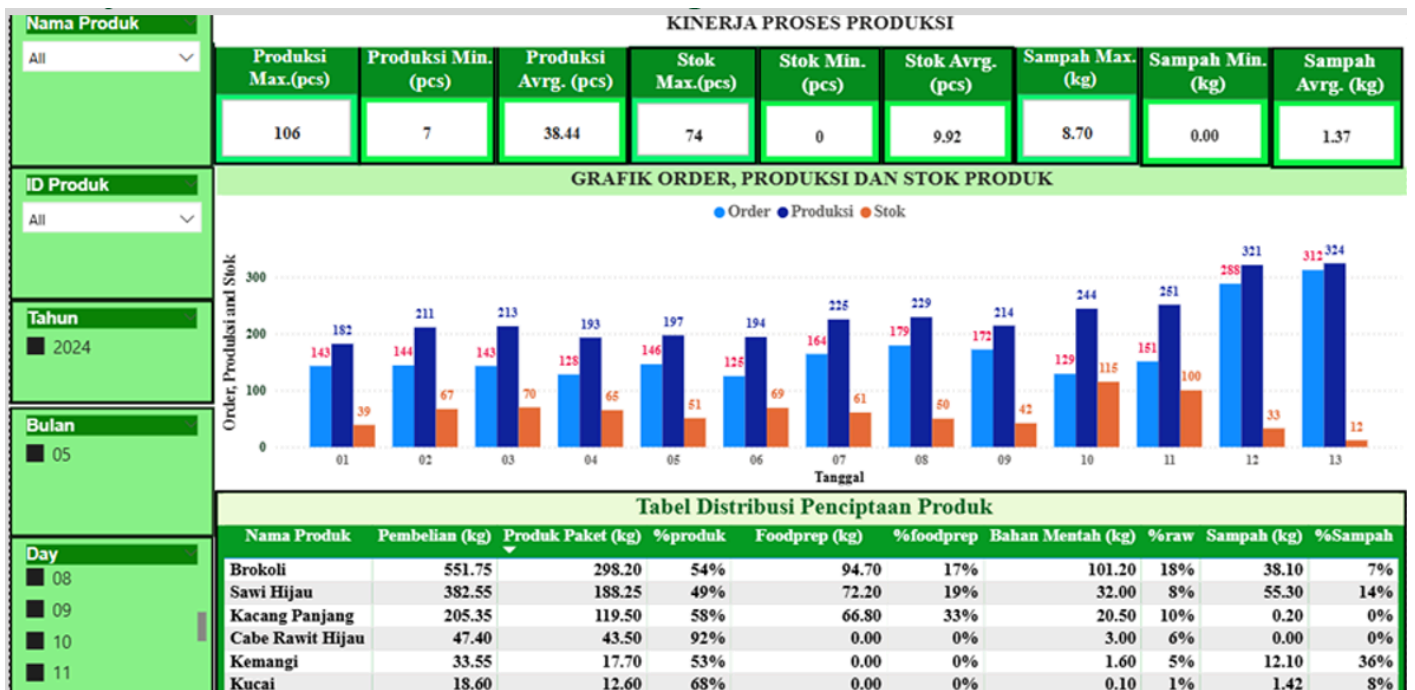
## Intelligen Bisnis ->ETL -->OLAP --- >Tabel Fakta Produksi

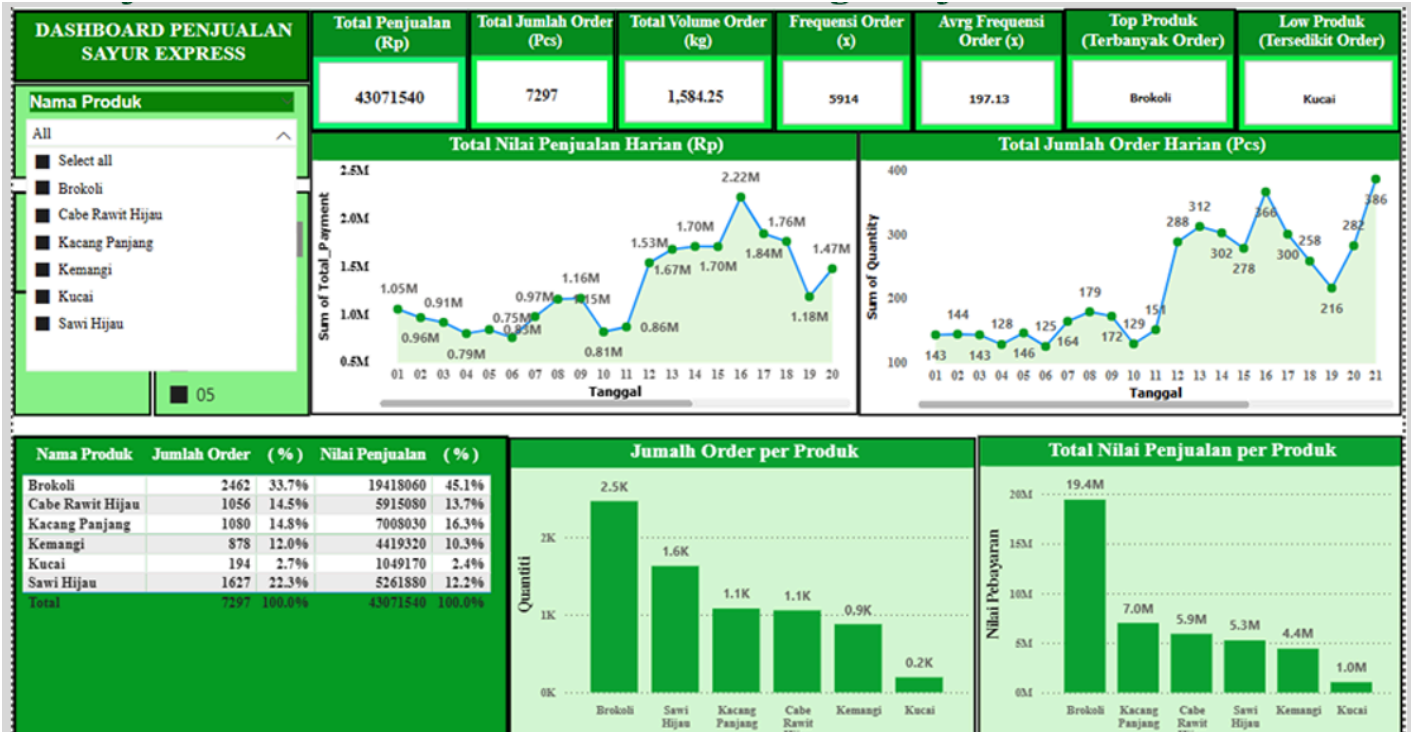


## Intelligen Bisnis ->ETL --> OLAP --- > Tabel Fakta Sisa Stok



7. Loading - visualisasi data dan informasi dengan Power BI





## Insight Informasi Dashboard

| Hambatan                                                                          | Solusi                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dashboard Penjualan</b>                                                        |                                                                                                                                                                                               |
| 1 Fluktuasi penjualan harian; adanya ketergantungan pada hari atau momen tertentu | Strategi promo yang lebih konsisten dan memanfaatkan media sosial dan WhatsApp untuk mengingatkan pelanggan.                                                                                  |
| 2 Dominasi produk spesifik                                                        | Tingkatkan variasi produk terlaris sesuai selera pelanggan.<br>Inovasi kemasan pada produk kurang laku.<br>Survei pelanggan untuk memahami preferensi produk.                                 |
| 3 Produk dengan penjualan rendah                                                  | Mengevaluasi produk dengan penjualan rendah.<br>Menggunakan strategi flash sale atau diskon khusus.<br>Membuat konten edukasi di media sosial tentang manfaat produk.                         |
| <b>Dashboard Order, Produksi , Stok</b>                                           |                                                                                                                                                                                               |
| 1 Produksi tidak seimbang dengan permintaan                                       | Sistem produksi berbasis peramalan permintaan<br>Sistem pre-order atau sistem keanggotaan sayuran                                                                                             |
| 2 Kelebihan stok pada produk tertentu                                             | Promo bundling atau diskon produk dengan stok berlebih<br>Jual cepat produk stok tinggi & umur simpan pendek                                                                                  |
| 3 Pesanan tidak stabil, menyebabkan produksi tidak konsisten                      | Perbaiki strategi pemasaran agar pesanan konsisten<br>Identifikasi kapan pelanggan berbelanja banyak / sedikit<br>Membangun loyalitas pelanggan agar bisa diprediksi belanja                  |
| 4 Stok Produk Rendah atau Nol                                                     | Membangun sistem prediksi permintaan & tingkatkan buffer stok<br>Pengadaan bahan baku lancar untuk produksi produk favorit<br>Sistem harga dinamis saat permintaan tinggi agar profit optimal |